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Descendants: Architecture, Embodiment, and Emergent Behavior in Browser-Based 3D Generative Agent Simulations

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DECLARATION

I hereby declare that the work presented in this thesis, entitled **"Descendants: Architecture, Embodiment, and Emergent Behavior in Browser-Based 3D Generative Agent Simulations"**, is my own original work, except where otherwise indicated and duly acknowledged. All sources of information and assistance used in the preparation of this thesis have been cited or referenced.

This thesis has not been submitted, either in whole or in part, for any other degree or qualification at this or any other institution. I understand that any deliberate misrepresentation or omission in this declaration may constitute academic misconduct and may lead to disciplinary action.

_____ _____

Signature of Candidate Date

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I also wish to acknowledge the Descendants open-source project community, whose collaborative engineering efforts provided the foundational platform infrastructure for this study. The commitment to open-sourcing all research-critical components under an Apache 2.0 license ensures the reproducibility and accessibility of this work.

Finally, I wish to thank my family and friends for their patience and encouragement during the preparation of this thesis, and my research team members for their collaborative spirit and dedication to academic rigour.

ABSTRACT

Large Language Model (LLM)-driven generative agents have demonstrated remarkable capabilities in simulating human-like behaviour within virtual environments. The transition to 3D spatial environments, while capable of unlocking enhanced realism and complex affordance processing, introduces significant new architectural and reasoning challenges that require novel systemic approaches.

This research introduces Descendants, a rigorous, open-source 3D browser-based simulation platform designed to investigate LLM-driven agent behaviour and system architecture in rich spatial environments. The study demonstrates the platform's capabilities by deploying generative agents (powered by Llama models via the Groq Cloud API) across experimental scenarios of increasing social complexity within a simulated 3D town.

The platform's architecture features a novel Inter-Agent Communication (IAC) library that decouples agent cognition from the underlying spatial engine. The system introduces a Neuro-Symbolic Agent Pipeline with symbolic verification, a Mixture-of-Experts (MoE) context mixer, and a utility-based subconscious engine that provides deterministic fallback behaviour.

By publishing the complete Descendants architecture, this study establishes a reusable, robust multi-agent research framework for 3D generative agent research.

Keywords: *Generative Agents, Large Language Models, Embodied Cognition, Multi-Agent Simulation, 3D Browser Environments, React Three Fiber, Spatial Reasoning*

LIST OF ABBREVIATIONS

LLM	Large Language Model
AI	Artificial Intelligence
NPC	Non-Player Character
R3F	React Three Fiber
IAC	Inter-Agent Communication
MoE	Mixture of Experts
API	Application Programming Interface
SDK	Software Development Kit
FPS	Frames Per Second
APA	American Psychological Association
OSF	Open Science Framework
ANOVA	Analysis of Variance
DOI	Digital Object Identifier
USJP	University of Sri Jayewardenepura
FOT	Faculty of Technology

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CHAPTER 1

INTRODUCTION

1.1 Background

Large Language Model (LLM)-driven agents have recently demonstrated remarkable capabilities in simulating human-like behaviour in virtual environments. These generative agents can follow daily routines, remember past interactions, form relationships, and coordinate with other agents to accomplish complex tasks (Park et al., 2023). The seminal "Generative Agents" work from Stanford University showcased these capabilities in Smallville, a text-driven environment where 25 agents lived and interacted over two simulated days.

While these demonstrations have been compelling, the research community has not yet explored how generative agents perform in architecturally complex 3D spatial environments. The transition to 3D environments introduces several critical challenges:

- Spatial Complexity: 3D spaces provide additional degrees of freedom for movement and positioning. Agents must reason about vertical space, occlusion, and volumetric relationships rather than simple planar coordinates.
- Perceptual Richness: 3D environments offer richer visual information, including depth cues, perspective, and more naturalistic representations of spatial relationships. This influences how agents perceive and reason about their surroundings.
- Navigational Challenges: Pathfinding in 3D requires more sophisticated algorithms (e.g., navmesh-based navigation), affecting agent movement patterns and spatial decision-making.
- Computational Overhead: 3D rendering and physics simulation impose greater computational costs, which influence system design choices and practical deployment scenarios.
- Description Complexity: When agents communicate about their environment or receive perceptual input, 3D spatial information requires different linguistic descriptions and increased token consumption when interfacing with LLMs.

From a theoretical perspective, embodied cognition theory (Shapiro, 2019) suggests that the physical form and sensorimotor capabilities of an agent shape its cognitive processes. If this principle extends to virtual embodiment, we can expect that 3D spatial embodiment produces qualitatively different and richer agent behaviours.

Understanding these challenges has practical implications for researchers and practitioners building multi-agent simulations, educational tools, demonstration platforms, and light-weight AI testbeds. Exploring 3D embodiment and demonstrating its viability justifies the additional engineering complexity and computational expense required for realistic simulations.

To investigate these questions, we developed the Descendants platform, an open-source, modular simulation engine built with modern web technologies. The platform features a novel Inter-Agent Communication (IAC) library that decouples agent cognition from spatial embodiment. Development has strategically prioritised the 3D implementation to validate the more technically demanding component of the system.

1.2 Scope of the Study

This study focuses on constructing and evaluating browser-based, visually grounded multi-agent simulations operating fully within a 3D architecture.

The technical specifications of the Descendants continuous 3D spatial simulation are as follows:

Technology	Representation	Pathfinding
React Three Fiber + Three.js + Yuka AI	3D capsule meshes in procedural space	A* on navmesh

Table 1: Technical specifications of the 3D embodiment architecture

1.2.1 Evaluative Scenarios

Three scenarios of increasing social complexity are instantiated within the 3D platform to test the robustness of the system:

1. Daily Routine Execution: Agents follow dynamic daily schedules in real-time, adapting pathfinding and goals based on environmental events.
2. Information Propagation: An agent receives novel information. System traces track the timeline of this propagation via agent-to-agent continuous communication.
3. Collaborative Planning: Agents coordinate shared goals involving spatial convergence, synchronised arrival, and resource allocation.

1.2.2 LLM Models

Model	Provider	Role
Llama 4 Maverick	Groq Cloud	Primary cognitive engine
Llama 3.1 8B	Groq Cloud	Fast fallback and lightweight tasks

Table 2: LLM models used in the system

1.2.3 Measured Outcomes

- Behavioural Metrics: Task success rates in 3D, observed instances of complex spatial reasoning, conversational logic bound to spatial facts.
- System Metrics: API action planning latency (ms), token consumption per cognitive cycle, frame rate (FPS) rendering impact.

1.2.4 Scope Limitations

Included	Excluded
Browser-based 3D simulation	High-fidelity physics (water, softbody)
Agent cognitive pipeline	Complex kinematic manipulation
Evaluation of system architecture	Large-scale environments (>50 agents)
Open-source framework	Human-in-the-loop interactions
Behavioural analysis in 3D	Extensive LLM cross-model evaluation

Table 3: Scope boundaries of the study

The deliberately constrained scope ensures project completion within an undergraduate research timeline while maintaining sufficient rigour for publishable findings on spatial agent architectures.

1.3 Research Problem

Despite growing interest in generative agents and LLM-driven multi-agent systems, there is a fundamental gap in systematic architectures that robustly bridge symbolic language processing with continuous 3D spatial execution. Existing foundational studies have relied on simplistic grid-based environments where spatial reasoning is trivially reduced to coordinate mapping.

The core problem driving this research is: How can a multi-agent cognitive architecture be designed to reliably scale emergent generative agent behaviour within a dynamic, continuous 3D spatial environment?

1.4 Research Questions

- RQ1, Systems Architecture: What are the systemic mechanisms and modular architectures (e.g., Context Mixers, Subconscious Engines) required to bridge symbolic LLM reasoning with continuous spatial navigation and physics?
- RQ2, Behavioural Robustness: Does the implementation of a 3D structural embodiment allow agents to overcome the spatial grounding errors prevalent in simpler text-based representations?
- RQ3, System Trade-offs: What are the computational and resource trade-offs inherent in driving continuous 3D simulations via high-latency LLM API calls, and how can they be mitigated?

1.5 Objectives

1.5.1 Main Objective

To design, develop, and evaluate Descendants, a structurally robust, open-source 3D browser-based simulation platform, exploring how a Neuro-Symbolic architecture can enable rich behavioral dynamics in continuous 3D environments.

1.5.2 Specific Objectives

1. Platform Engineering: Implement the fully functioning Descendants continuous 3D spatial environment with procedural navigation and deterministic Subconscious Engines.
2. Agent Cognitive Library: Develop the `Descendants IAC` library enabling LLMs to natively process 3D Spatial and VisualCortex context.
3. Behavioural Deployment: Deploy agents with distinct personae into the simulation and capture their interactions during predefined scenarios.
4. Benchmarking: Evaluate qualitative and quantitative outcomes (such as spatial error rates and coordination complexity) of the 3D generative agent architecture.
5. Reproducibility and Open Science: Release a complete open-source experimental framework under an Apache 2.0 licence, providing a modular basis for future 3D embodiment research.

CHAPTER 2

LITERATURE REVIEW

2.1 Generative Agents and Social Simulation

The foundation for this research lies in the "Generative Agents" paradigm introduced by Park et al. (Park et al., 2023). Their work demonstrated that LLMs could power agents with emergent social behaviours through a memory, reflection, and planning architecture. Twenty-five agents inhabited a virtual town, autonomously forming relationships, spreading information, and coordinating activities, all driven by a combination of LLM reasoning and a structured memory system.

This research builds directly on their contribution and extends it into fully 3D continuous spatial environments, exploring the architectural requirements for deploying generative agents with true volumetric embodiment.

Subsequent frameworks have extended the multi-agent paradigm in different directions. AgentVerse (Chen et al., 2023) facilitates multi-agent collaboration through task-oriented group dynamics, while CAMEL (Li et al., 2023) explores communicative agents through structured role-playing dialogues. AutoGen (Wu et al., 2023) provides a framework for building multi-agent conversational systems. However, these frameworks focus primarily on disembodied, task-oriented coordination, leaving the role of spatial grounding in open-ended social simulation underexplored.

2.2 Embodied AI Platforms

In parallel with the generative agent research, the embodied AI community has developed sophisticated 3D platforms for studying agent behaviour in physical environments. Habitat (Savva et al., 2019) provides a high-performance 3D simulation platform for training embodied agents in photorealistic indoor environments. The platform has been instrumental in advancing research on visual navigation, object interaction, and embodied question answering.

However, these platforms differ fundamentally from generative agent research in several ways. Habitat and similar platforms (AI2-THOR, iGibson) primarily use reinforcement learning for low-level sensorimotor tasks such as point-goal navigation, rather than language-driven social interaction.

Our work bridges this divide by applying the rigour of embodied AI research to the high-level social simulation goals of generative agent research, creating a platform that supports both paradigms within a continuous 3D environment.

2.3 Spatial Reasoning in Large Language Models

An LLM's ability to reason about space is central to this study. Recent work has demonstrated that LLMs possess some capacity for spatial reasoning from purely textual descriptions, but this capacity is limited and often relies on memorised heuristics rather than genuine spatial understanding.

The Voyager project (Wang et al., 2023) showed that interactive, embodied experience in Minecraft significantly improves agent performance on spatial tasks compared to static prompting. This finding suggests that the quality and structure of spatial information could fundamentally alter an LLM's reasoning process and subsequent behaviour.

In 3D environments, spatial information must account for depth, elevation, occlusion, and perspective (e.g., "Bob is standing on the balcony above you, partially obscured by a railing"). This richer descriptive requirement engages more sophisticated spatial reasoning in the LLM, potentially leading to more nuanced behavioural outcomes.

2.4 Multi-Agent Communication Frameworks

Research on multi-agent communication has progressed along two parallel tracks. Frameworks like MetaGPT (Hong et al., 2023) show that LLM-driven agents can collaborate effectively through structured linguistic protocols, assigning roles such as product manager, architect, and engineer to coordinate complex software development tasks. However, these interactions are entirely disembodied, with agents communicating through text channels without spatial context.

In contrast, multi-agent reinforcement learning research demonstrates that agents develop emergent communication protocols that are deeply grounded in their shared physical space. The properties of the environment, including dimensionality, visibility, and proximity, directly shape the nature and efficiency of these protocols.

Our study investigates how the pre-trained linguistic abilities of LLMs are modulated by 3D spatial contexts, potentially leading to richer emergent communication and coordination strategies natively situated in fully continuous environments.

2.5 Evaluation Methodologies for Agent Systems

Evaluating emergent behaviour in multi-agent systems is notoriously challenging. The original Smallville study used subjective "believability" ratings from human evaluators, which

provide ecological validity but lack precision and reproducibility. Embodied AI research uses objective task-completion metrics, but these can miss the nuance of social interaction.

We adopt a hybrid evaluation approach, triangulating multiple measurement modalities:

- Objective quantitative metrics: task success rates, information propagation speed, and system performance measurements.
- Semi-objective rubric-based coding: trained raters score coordination quality using a standardised 5-point rubric, with inter-rater reliability assessed via Cohen's Kappa.
- Qualitative dialogue analysis: thematic analysis of agent-generated dialogues to examine spatial language use and coordination strategies.

This multi-modal approach ensures that both measurable outcomes and qualitative behavioural patterns are captured.

2.6 Gap Identification and Research Contribution

The literature reveals a critical gap: no systematic 3D architecture for deploying open-ended, continuous, multi-agent generative agents has been fully implemented and evaluated. Existing work either operates in simplified environments or lacks the Neuro-Symbolic integration needed for naturalistic spatial interactions.

Our research addresses this gap through three primary contributions:

1. Providing the first structural 3D platform capable of orchestrating autonomous, LLM-driven agents seamlessly.
2. Releasing a reusable open-source framework (the Descendants platform and IAC library) to set a new benchmark for spatial generative agents.
3. Contributing to a theoretical understanding of virtual embodiment for AI agents, extending embodied cognition theory to the domain of continuous LLM-driven simulacra.

CHAPTER 3

THEORETICAL FRAMEWORK

3.1 Embodied Cognition Theory

Our theoretical foundation draws primarily from embodied cognition theory, which posits that cognitive processes are deeply rooted in the body's interactions with the world (Shapiro, 2019) (Wilson, 2002).

3.1.1 Core Principles

- Situatedness: Cognition occurs in real-time context. Mental representations are grounded in sensorimotor experience rather than existing as abstract, disembodied symbols.
- Action-Oriented Representation: Knowledge is structured not for passive storage but to guide action. Spatial knowledge, for example, is encoded as affordances (possibilities for action) rather than abstract coordinates.
- Cognitive Offloading: Intelligent systems use the environment itself to reduce internal computational load. Agents in richer environments may offload more reasoning to environmental cues.
- Body-Specific Constraints: An agent's physical form, its perceptual apparatus, action repertoire, and spatial extent, shapes and constrains its cognitive processes.

3.1.2 Extension to Virtual Embodiment

We extend these principles to virtual agents, where embodiment manifests through three channels:

1. Perceptual input format: how the 3D environment is described to the LLM via spatial narratives.
2. Available action space: the set of actions the agent can perform and how they map to spatial transformations.
3. Environmental feedback: how the environment responds to agent actions and presents new information.

We hypothesise that the spatial layout and complexities of 3D semantic information inherently influence the computational load and descriptive accuracy required. As established by embodied theories, continuous depth and dynamic occlusion introduces measurable variance in the LLM's spatial logical execution.

3.2 Spatial Affordances and Agent Behaviour

Building on Gibson's (Gibson, 1979) theory of affordances, we analyse how 3D environments offer possibilities for action and perception.

3.2.1 Navigational Affordances

In 3D environments, agents must reason about vertical space, volumetric occlusion, and perspective-dependent visibility. Navigational decisions require reasoning about elevation, line-of-sight, and spatial relationships that introduce significant architectural complexity compared to simple coordinate systems.

3.2.2 Perceptual Affordances

Agents perceive from an egocentric perspective. Perceptual information must account for line-of-sight and occlusion (e.g., "You see Bob standing across the square. A tree partially obscures your view."). This richer descriptive requirement engages sophisticated spatial reasoning in the LLM.

3.2.3 Social Affordances

3D environments afford complex social phenomena: gaze direction, joint attention (looking at the same object), eavesdropping from behind cover, and a clearer sense of personal space and privacy through occlusion.

3.3 Hypothesis Development

Based on our theoretical framework, we develop the following testable hypotheses:

Hypothesis	Statement
H1 (Architectural Viability)	The proposed Descendants Neuro-Symbolic architecture, via its IAC library and Context Mixer, will successfully drive coherent agent behaviours in continuous 3D environments.
H2 (Spatial Language)	Dialogues and internal thoughts generated by 3D agents will contain a high frequency of complex spatial prepositions (e.g., “behind,” “under,” “through”) and perspective-based language.
H3 (System Cost)	Continuous 3D simulation will incur measurable computational costs, measured by action planning latency and token consumption per decision cycle, due to the complexity of describing and reasoning about 3D space.
H4 (Model Invariance)	The direction and significance of effects observed in H1 to H3 will be consistent across both Llama 4 Maverick and Llama 3.1 8B, suggesting embodiment effects are fundamental rather than model-specific artefacts.

Table 4: Research hypotheses and their theoretical basis

CHAPTER 4

METHODOLOGY

4.1 Research Design

This study employs an Exploratory Systems-Building and Scenario-Based Evaluation design. The research evaluates the capabilities of the Descendants continuous 3D multi-agent architecture across varied scenarios and cognitive engines:

Factor	Levels
Architecture	Descendants 3D Platform
Large Language Model	Llama 4 Maverick vs. Llama 3.1 8B (via Groq Cloud)

Table 5: Evaluation Design Structure

4.2 Experimental Scenarios

Scenario	Description	Success Metric
A, Daily Routine	Agents follow individual schedules across a simulated day in 3D space (wake, work, socialise, sleep), adapting to environmental conditions and dynamic pathing.	Percentage of schedule items completed without spatial breakdown or pathing failure.
B, Information Propagation	One agent receives novel information at simulation start. We measure how the information spreads through organic social interactions given spatial occlusion.	Simulated minutes until 100% of agents have been informed via spatial proximity conversations.
C, Collaborative Planning (Surprise Party)	Agents must collectively plan a surprise birthday party. Requires spatial convergence, role allocation, and secrecy.	-

Table 6: Experimental scenarios with success metrics

4.3 Variables and Operationalisation

4.3.1 Independent Variables

- `LLM Engine`: Categorical: {Llama 4 Maverick, Llama 3.1 8B}. Manipulated through the Groq API endpoint configuration to test cognitive reasoning impact.
- `Scenario Type`: Categorical: {Routine, Propagation, Collaboration}.

4.3.2 Dependent Variables

- `TaskSuccess`: Ratio/Interval scale. Operationalised as the scenario-specific success metric.
- `CoordinationQuality`: Ordinal scale (1 to 5). Rated by trained human coders using the Coordination Quality Rubric (CQR).
- `SpatialLanguageFrequency`: Ratio scale. Count of spatial prepositions and perspective-based language per dialogue turn.
- `Latency`: Ratio scale. Milliseconds from observation to executed action.
- `TokenConsumption`: Ratio scale. Total tokens consumed per agent per decision cycle.

4.3.3 Control Variables

- `RandomSeed`: Fixed per run to ensure deterministic agent scheduling.
- `PromptTemplates`: Identical system prompts and role descriptions across conditions.
- `EnvironmentLayout`: Consistent 3D environmental configuration verified through the Parity Verification Protocol.

4.4 Data Collection Procedures

All agent actions, perceptions, internal thoughts, communications, and system performance metrics are automatically logged to a structured JSON database. Each log entry is time-stamped and tagged with the agent ID, run number, and experimental condition. The logging system captures:

- Perception logs: The full text prompt sent to the LLM at each decision cycle, including spatial context and memory.
- Decision logs: The LLM's response, including tool calls, reasoning text, and selected action.
- Action logs: The executed action, verification results, and any symbolic fallback activations.
- System logs: Latency, token counts, frame rate, and memory usage.

Data collection is fully automated to prevent manual intervention or bias.

4.5 Measurement Instruments

4.5.1 Coordination Quality Rubric (CQR)

Two trained, independent raters blind to the experimental condition score a random 20% sample of interaction logs from Scenario C using the following 5-point scale. Inter-rater reliability is assessed using Cohen's Kappa (target $k > 0.7$).

Score	Label	Description
1	Chaotic	Agents act independently; actions are contradictory or counter-productive.
2	Reactive	Agents respond to others but do not proactively plan together.
3	Coordinated	A shared plan is evident, but execution is inefficient or incomplete.
4	Collaborative	Agents explicitly allocate roles and synchronise actions effectively.
5	Synergistic	Agents dynamically adapt roles, resolve conflicts, and leverage each other's capabilities.

Table 7: Coordination Quality Rubric (CQR) scale

4.6 Statistical Analysis Plan

All analyses are conducted using R with a significance level of a = 0.05.

- A one-way ANOVA is used to analyse the main effect of LLM model on each continuous dependent variable (TaskSuccess, SpatialLanguageFrequency, Latency, TokenConsumption).
- For the ordinal `CoordinationQuality` data, the non-parametric Mann-Whitney U test is used.
- Effect sizes are reported using partial eta-squared for ANOVA and rank-biserial correlation for non-parametric tests.
- Post-hoc pairwise comparisons use Tukey's HSD with Bonferroni correction where applicable.

4.7 Qualitative Analysis Protocol

In addition to quantitative analysis, we perform a thematic analysis of dialogue excerpts from the highest- and lowest-performing runs in Scenario C. Two researchers independently code the dialogues using an open coding approach, followed by axial coding to identify themes related to spatial language use, coordination strategies, and environmental reference. This analysis provides illustrative examples and generates qualitative insights into how and why coordination succeeds or fails.

CHAPTER 5

TECHNICAL ARCHITECTURE

5.1 System Overview

The system is architected in a modular fashion, cleanly separating the agent's "mind" (LLM-based reasoning) from the environment's "body" (3D simulation). The Descendants platform serves as the foundational infrastructure, with the research-specific modules organised into three core layers:

1. Descendants IAC Library (`@descendants-iac/core`, `@descendants-iac/react`): A framework-agnostic agent orchestration layer providing the Agent class, CapabilityAdaptor system, SpatialOracle, and MemoryContext.
2. World Engine: The 3D simulation environment built with React Three Fiber, Three.js, and Yuka AI for spatial rendering, physics, and autonomous navigation.
3. Implementation Harness: Logging, data collection, and scenario execution infrastructure.

All research-critical components are released under an Apache 2.0 licence.

5.2 Platform Specification

Component	Specification
Core Platform	Descendants v2.3.1
Framework	Next.js 16 (React 19)
Language	TypeScript
3D Rendering	React Three Fiber v9, Three.js r181
AI/ML	Vercel AI SDK, Groq Cloud API (@ai-sdk/groq)
LLM Models	Llama 4 Maverick, Llama 4 Scout, Llama 3.3 70B, Llama 3.1 8B
Autonomous Navigation	Yuka AI v0.7.8
State Management	Zustand v5
Schema Validation	Zod v4
Determinism	Deterministic mode enabled via fixed seeds

Table 8: Platform technology stack

5.3 Agent Architecture: The Neuro-Symbolic Pipeline

The agent architecture follows a Perceive, Plan, Act cycle, implemented through a Neuro-Symbolic Pipeline that combines LLM-based reasoning with deterministic symbolic verification.

5.3.1 The Agent Class

The `Agent` class (from `@descendants-iac/core`) is the central orchestrator for each autonomous entity. It manages:

- Identity and Role: System prompt construction from agent ID and role description.
- Memory: A rolling `MemoryContext` with configurable short-term history window.
- Capabilities: An `AdaptorRegistry` that converts registered capabilities into Vercel AI SDK tool definitions.
- Spatial Awareness: A `SpatialOracle` that generates text-based spatial context from the 3D world state.
- Subconscious State: `AgentNeeds` (energy, social, curiosity, stress) and `PersonalityMatrix` (sociability, diligence, neuroticism, curiosity) that decay and accumulate over time.

Each agent “tick” follows a five-stage pipeline:

1. Subconscious Tick: Update internal needs based on time decay and personality modifiers.
2. Perception: The SpatialOracle compiles spatial context; the MemoryContext assembles the rolling message history.
3. LLM Reasoning: The compiled messages and available tool schemas are sent to the LLM via the Vercel AI SDK.
4. Symbolic Verification: The returned action is verified against spatial preconditions using the CapabilityAdaptor's `verify()` function.
5. Execution or Rejection: Valid actions proceed to the CapabilityEngine; invalid actions are stored as failed tool calls for the LLM to learn from.

5.3.2 The CapabilityAdaptor System

The `CapabilityAdaptor` interface defines a single agent action with five aspects:

- Schema (`name`, `description`, `parameters`): Defines the action for the LLM using Zod schemas, automatically converted to Vercel AI SDK tool definitions.
- Symbolic Verifier (`verify`): Validates preconditions before execution (e.g., "Can the agent walk to this target? Is the target within range?").
- Controller (`execute`): Physical execution logic that interfaces with the game engine (Yuka steering, Three.js transformations).
- Reflector (`reflect`): Post-execution metacognitive evaluation of the action outcome.
- Utility Scorer (`baseUtilityScore`, `utilityCondition`): Subconscious scoring based on agent needs and personality for deterministic fallback behaviour.

Built-in capabilities include: `idle`, `walk`, `explore`, `sit_on_edge`, `pick_up_item`, `drop_item`, `navigate_to_anchor`, `navigate_to_coord`, `follow`, and `social_interact`.

5.3.3 The Subconscious Engine (Utility Evaluator)

When the LLM is unavailable (rate limiting, network failure, or high latency), the `UtilityEvaluator` provides deterministic fallback behaviour. It scores each registered capability based on:

$$\text{FinalScore} = \text{baseUtilityScore} \times \text{utilityCondition}(\text{needs}, \text{personality}, \text{spatial}) \quad (1)$$

The highest-scoring valid action (verified symbolically) is executed automatically, ensuring agents never "freeze" even when LLM calls fail.

5.4 Context Mixer: Mixture-of-Experts Architecture

The Context Mixer assembles the agent's perceptual context using a Mixture-of-Experts (MoE) architecture, where each "expert" (called a Shard) handles a specific aspect of cognition:

5.4.1 Spatial Shard

The Neuro-Symbolic bridge. Uses the `YukaOracle` to provide physics and navigation context, including current zone, nearest landmarks, escape routes, and cover positions. This shard resolves continuous 3D coordinate space into structured semantic context.

5.4.2 Visual Cortex Shard

Simulates visual perception with a configurable field-of-view (default: 120 degrees) and maximum range (20 units). Filters perceived entities to the top 5 by relevance, providing direction and distance information. This shard includes vertical positioning and occlusion data.

5.4.3 Hippocampus Shard

Manages episodic memory with importance-weighted storage. Recent memories are maintained in a sliding window, with periodic consolidation of older memories into summary entries. Each memory is typed (action, observation, conversation, summary) and timestamped.

5.4.4 Social Shard

Tracks inter-agent relationships with familiarity scores, sentiment values, interaction counts, and recency. Generates social context for the LLM including current interaction partner and known relationship history.

5.4.5 Amygdala Shard

Models emotional state along two dimensions: valence (pleasant/unpleasant, range 0 to 1) and arousal (calm/excited, range 0 to 1). Emotional state modulates the agent's tendency towards social engagement, risk-taking, and stress responses.

Each shard provides a `getRelevance(trigger)` method that returns a weight based on the current trigger type (TIMER, PLAYER_NEARBY, AGENT_NEARBY, HURT, IDLE). The Context Mixer selects the most relevant shards to include in the LLM prompt, managing token budgets intelligently.

5.5 Environment Implementation

The 3D world engine is fully implemented using React Three Fiber and provides:

- Dynamic Time System: Continuous day/night cycle with accurate sun/moon positioning and atmospheric coloring.
- Procedural Terrain: FBM noise-based terrain with instanced foliage.
- Multi-Zone Architecture: Distinct environmental zones (residential, commercial, social hub, office) with a `ZoneController` managing smooth fog, lighting, and ambient transitions.
- Autonomous Navigation: Yuka AI `Vehicle` entities with `SeekBehavior`, `WanderBehavior`, and `SeparationBehavior` for crowd dynamics. The

`CapabilityEngine` handles navigation to named anchors via the `WorldRegistry`, with hysteresis-based arrival detection.

- Entity System: `AIRobot` components managing mesh, animation, interaction UI, and AI controller hooks.
- Camera System: Third-person camera rig with smooth damping and target locking.

5.6 Instrumentation and Logging

The system captures every perception prompt, LLM response, executed action, and system metric (latency, token count) for every agent at every time step. Logs are stored in structured JSON format with the following schema per entry:

- `timestamp`: Simulation time
- `agent_id`: Unique agent identifier
- `run_id`: Scenario deployment identifier
- `perception`: Full perception prompt text
- `response`: LLM response text and tool calls
- `verification`: Symbolic verification result
- `execution`: Action executed and outcome
- `metrics`: Latency (ms), token count, FPS

CHAPTER 6

REPRODUCIBILITY AND OPEN SCIENCE

6.1 Pre-registration Plan

Before any data collection commences, this study will be pre-registered on the Open Science Framework (OSF). The pre-registration includes:

- The full research proposal with hypotheses, experimental design, and statistical analysis plan.
- Exact scenario specifications and agent configurations.
- The Coordination Quality Rubric (CQR) and coding instructions.

Pre-registration ensures that our analysis is not influenced by observed results and commits us to reporting all findings, including null results.

6.2 Open Source Release Strategy

All research-critical source code is released on a public GitHub repository under an Apache 2.0 licence. The release follows a phased approach:

Month	Release
Month 1	Initial release of the Descendants IAC library (`@descendants-iac/core`, `@descendants-iac/react`), API definitions, and scenario specifications.
Month 3	Alpha release of the 3D environment templates, agent runtime, and experiment harness.
Month 6	Final release containing all experiment code, analysis scripts, raw data, and documentation.

Table 9: Open source release timeline

6.3 Data and Code Availability

Upon completion, the following will be made publicly available:

- GitHub Repository: All source code for the Descendants platform, IAC library, experiment harness, and analysis scripts.
- OSF Project: Pre-registration document, raw and processed experimental data, analysis scripts, supplementary materials.
- Zenodo Archive: Dataset and code snapshot with a permanent DOI for stable citation.

6.4 Documentation Standards

The open-source code is accompanied by comprehensive documentation, including:

- A `README.md` with setup instructions and system requirements.
- API documentation for the Descendants IAC library.
- A step-by-step tutorial demonstrating how to configure and run a sample experiment.
- Inline code documentation following TypeScript JSDoc conventions.

CHAPTER 7

TIMELINE AND MILESTONES

7.1 Project Timeline

Month	Phase	Key Activities and Milestones
Month 1	Setup and Foundation	Finalise proposal and secure approvals. Complete Ethics filings. Milestone: Pre-register study on OSF; release initial Descendants IAC library.
Month 2	Implementation	Complete 3D environment implementation. Implement full agent runtime and LLM integration. Milestone: Complete Parity Verification Protocol.
Month 3	Pilot and Refinement	Implement all three experimental scenarios. Conduct pilot study (5 runs per condition). Milestone: Validate measurement instruments (CQR k > 0.7).
Months 4 to 5	Data Collection	Run full set of experimental sessions. Monitor automated data collection and logging integrity. Milestone: Complete data collection; verify data integrity.
Month 6	Analysis and Write-up	Execute pre-registered statistical analysis. Perform qualitative thematic analysis. Milestone: Final thesis submission and complete open-source release.

Table 10: Project timeline across six months

7.2 Risk Management

Risk	Likelihood	Impact	Mitigation Strategy
Technical Delays	Medium	High	Modular architecture allows incremental development; simplified fallback scenario available.
LLM API Changes	Low	High	Use versioned API endpoints; cache all LLM responses to ensure reproducibility and control costs.
Null Results	Medium	Medium	A null result is still a valuable scientific finding; will be reported as committed in the pre-registration.
Exceeding Budget	Low	Medium	Token consumption closely monitored during pilot study to refine cost estimates.
Inter-rater Disagreement	Medium	Medium	Extensive rater training; pilot coding session; clear rubric with examples; fallback to third rater.

Table 11: Risk assessment and mitigation strategies

7.3 Deliverables

1. A final thesis document submitted to the University of Sri Jayewardenepura Faculty of Technology.
2. A public, open-source GitHub repository containing the Descendants platform, IAC library, and experiment code.
3. A public dataset on OSF with raw experimental data.
4. A manuscript for submission to a peer-reviewed academic conference (e.g., AAMAS, AAAI, CHI).

CHAPTER 8

BUDGET AND RESOURCES

8.1 Budget

8.1.1 Estimated Token Usage

The experimental design requires approximately:

$$120 \text{ runs} \times 30 \text{ min/run} \times 20 \text{ cycles/min} \times 1500 \text{ tokens/cycle} \approx 54 \text{ Million tokens} \tag{2}$$

Model	Rate	Estimated Cost
Llama 4 Maverick (via Groq Cloud)	Free tier	~$0 USD
Llama 3.1 8B (via Groq Cloud)	Free tier	~$0 USD
Contingency (paid tier overflow)		~$10 USD
Total Estimated		~$10 USD

Table 12: Estimated LLM API costs

This budget is minimal as the Groq Cloud free tier provides sufficient capacity for the experimental runs. Additional allocation may be made if supplementary models are included for broader validation.

8.2 Resources

Resource	Details
Personnel	3 undergraduate researchers, 2 academic supervisors
Hardware	Personal laptops (~16 GB RAM); university computing labs available if needed
Software	All open-source: React Three Fiber, Three.js, Yuka AI, Node.js, R
Platform	Descendants open-source simulation platform
AI Services	Groq Cloud API (free tier, Llama models)

Table 13: Resource allocation

CHAPTER 9

EXPECTED CONTRIBUTIONS

9.1 Empirical Contribution

This study provides rigorous empirical evidence on how continuous 3D embodiment affects the behaviour of LLM-driven generative agents. By deploying complex cognitive agents into rich spatial scenarios, we demonstrate the specific effects of continuous spatial constraints on agent decision-making, coordination, and communication patterns.

9.2 Methodological Contribution

The research delivers a standardised methodology and open-source framework for conducting multi-agent systems research in continuous 3D environments. The Descendants IAC library introduces a reusable architecture, the CapabilityAdaptor pattern with symbolic verification and utility-based fallback, that can be adapted for studying different architectural variables:

- Different LLM models
- Different memory architectures
- Different social structures
- Different environmental complexity levels

9.3 Theoretical Contribution

The findings contribute to an understanding of virtual embodiment for AI agents, extending embodied cognition theory (Shapiro, 2019) to the domain of LLM-driven simulation. Specifically, the study tests whether the core principles of situated cognition and body-specific constraints apply to agents whose “embodiment” is mediated entirely through text descriptions of spatial environments.

9.4 Technical Contribution

The Descendants platform and IAC library represent significant engineering contributions:

- Neuro-Symbolic Agent Pipeline: Combining LLM reasoning with deterministic symbolic verification to prevent hallucinated actions.
- Mixture-of-Experts Context Mixer: A modular, extensible architecture for assembling agent perception from heterogeneous cognitive shards.

- Utility-Based Subconscious Engine: Ensuring continuous agent operation even during LLM unavailability.

9.5 Community Contribution

All code, data, and materials are released publicly under an Apache 2.0 licence, enabling replication, extension, and education across the multi-agent systems, AI, and computational social science communities. The phased release strategy ensures the community benefits from our work at every stage of the project.

CHAPTER 10

CONCLUSION

This thesis outlines a focused and rigorous implementation strategy to address a fundamental architectural challenge in the field of generative agents: how to architect structurally robust, continuous 3D environments that interact with symbolic LLM cognition.

By systematically deploying and observing functionally equivalent agents within a comprehensive 3D environment, this project moves beyond anecdotal demonstrations to provide robust empirical metrics. The research is grounded in embodied cognition theory and operationalised through the Descendants multi-agent platform.

The Descendants platform, developed as the research infrastructure for this study, represents a significant technical contribution in its own right. Its novel architecture, featuring a Neuro-Symbolic Agent Pipeline with symbolic verification, a Mixture-of-Experts context mixer, and a utility-based subconscious engine, addresses fundamental challenges in building reliable, autonomous LLM-driven agents for spatial environments. The accompanying Descendants IAC library provides a reusable, engine-agnostic framework for agent orchestration that can serve as a foundation for future research.

Development has strategically prioritised a rigorous 3D implementation, recognising it as a technically demanding frontier for generative agent research. The 3D world engine, with dynamic time systems, procedural terrain, multi-zone architecture, and autonomous Yuka-based navigation, is fully operational.

The study is designed with a strong emphasis on academic integrity, featuring:

- A commitment to pre-registration on the Open Science Framework.
- Open-source release of all code under Apache 2.0.
- Public datasets with persistent DOIs for reproducibility.
- A feasible timeline for undergraduate research.

The expected contributions, empirical, methodological, theoretical, and technical, have the potential to significantly inform the future design and deployment of multi-agent simulacra. Whether the findings reveal spatial breakdown due to continuous complexity or demonstrate flawless spatial translation, the results will provide actionable guidance for researchers and practitioners making 3D embodiment design choices. We are confident that this work will yield valuable insights and provide a lasting resource for the open-source research community.

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